How warmer days make plant biology less predictable

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Overview

- A major concern with anthropogenic climate change is that increasing warming will alter biological processes, making forecasting, mitigation, and adaptation more difficult (1).
- Plant phenology—the timing of recurring life history events, such as leafout and flowering—has been routinely cited as an example, with a growing literature claiming that in response to warming, plants have slowed down, become more variable, or less predictable (2; 3)
- Here we present new findings using the theory of stopped random walks to show why spring events can become more variable with continued warming.
- Using datasets from across the Northern hemisphere, we find support for this result.
- Our findings suggests many biological events that depend on a thermal sum model may become less predictable with continued warming, with major implications for crops, wild plants and animals.

Background

- Plant phenological events in the spring, such as leafout and flowering of trees and shrubs, are triggered by accumulated warmth (forcing).
- Because of this thermal sum model (of which growing degree days is a common example) plant events happen more quickly at warmer temperatures in both controlled experiments (Figure 1) and in recent decades with anthropogenic warming.
- This thermal sum model is non-linear on an untransformed scale (Figure 1), which may explain recent slow-downs in advances of spring phenology (4).

Thermal sum model as stopped random walk

- The thermal sum (growing degree day) model is well-described by the theory of stopped random walks.

- Based on this model (using analytical solutions to the model, not shown), variance in spring events depends critically on whether temperatures accumulate during periods of relatively flat temperatures or during periods when temperatures rise linearly (Figure 2).
- During periods when temperatures are relatively flat (e.g. most temperate climates in the winter or experiments with constant 'forcing'), spring events have high variance as the error of the model plays a large role in determining the timing.
- During periods when temperatures are approximately linear (e.g. most temperate climates in the spring), the role of error in the model is dramatically reduced and spring events should show very low variance.

Desynchronizing plant events

- The thermal sum model thus may explain why cherry blossoms (and other spring events) across different microclimates bloom near simultaneously, as it suggests such effects may be negligible, but the model also predicts that variance may increase with warming.
- As temperatures warm, blooms may happen in the flatter temperature periods (Figure 2) where average temperatures do not increase and microclimate variation is not negligible, resulting in the desynchronization of blooms or a 'scattered spring.'

Increasing variance in spring events

Supporting this theory from stopped random walks, we tested for shifts in variance across different datasets of spring events.

We found increasing variance with warmer temperatures in all datasets we studied (Figure 3).

Using a very large dataset of lilacs from across the United States (5; 6), we estimated the shift in variance independent of the changing mean temperature using a generalized additive model (GAM), and found increasing variance as mean February temperatures pass 12 °C.

Global implications

As many biological events are triggered by accumulated exposure, our results have wide-reaching implications.

- We show that, under normal conditions, the model predicts both a slow down in advances of spring events (so-called 'declining sensitivity', 4; 7) and declining variance with warming.
- However, we also show that excessive warming can produce a new normal in which the variance actually increases.
- The data confirm that climate change is making many biological systems less predictable, with consequences for crops, carbon sequestration and thus climate change itself.

– Further, decades of experiments focused on maintaining controlled constant temperatures may fail to predict shifts in variance with increasing temperatures.

Figure captions

1. Controlled experiments across a wide temperature range show a non-linear response to temperature; data from **(author?)** (8), which reported a growth chamber experiment of tree branch cuttings of walnut trees (*Juglans regia*).

2. The underlying temperature regime is critical to predicting the variance of spring events with warming. Here we show simulated daily temperatures contrasting flat regimes (common in winter months) vs. linear regimes (common in spring months). Lines show mean (dashed) + AR(1) noise (grey).

3. Results showing earlier blooms (top) and increasing variance (bottom) with warming. We use February for consistency across datasets (other months produce similar results), splitting the years of data into three quantiles of temperatures, and estimating the standard deviation of event dates and its errors for each quantile. Cherry blossom data come from the US National Park service (see https://competition.statistics.gmu.edu/); UK oak leafout taken from **(author?)** (9); German beech data are from PEP725 (10), using the subset in **(author?)** (4).

4. Generalized additive model (GAM) results using lilac bloom data from across the United States (6).

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